

DECENTRALIZED NOTARY SMART CONTRACT

Remix IDE – Test Evidence

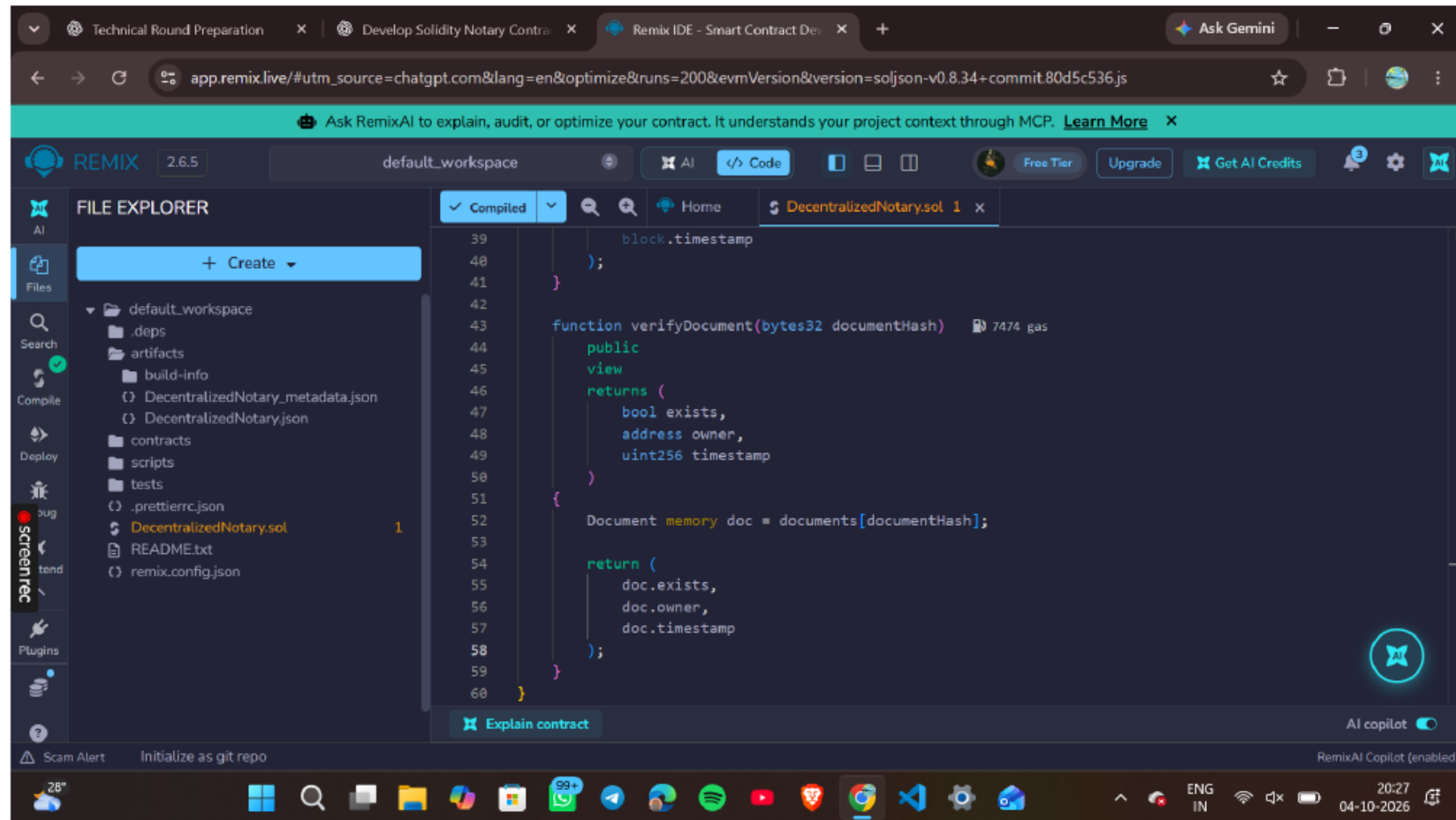
This document contains the implementation and testing evidence for the Decentralized Notary Smart Contract developed using Solidity and Remix IDE.

Project	Decentralized Notary Smart Contract
Language	Solidity ^0.8.20
Development Tool	Remix IDE
Execution Environment	Remix VM (Cancun)
Core Functions	registerDocument(), verifyDocument()

Prepared for academic project submission

1. Solidity Source Code

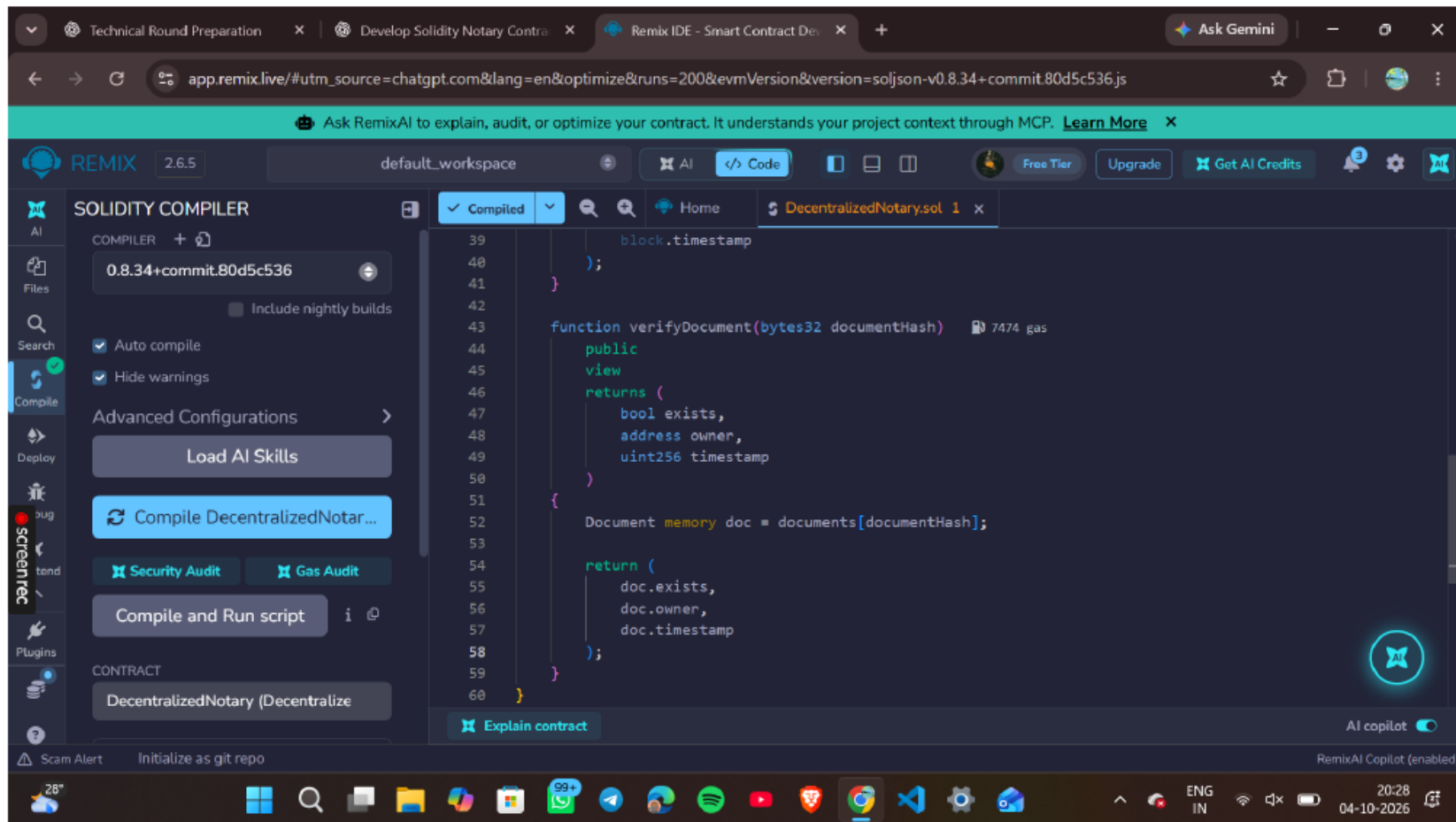
DecentralizedNotary.sol source code shown in Remix IDE.



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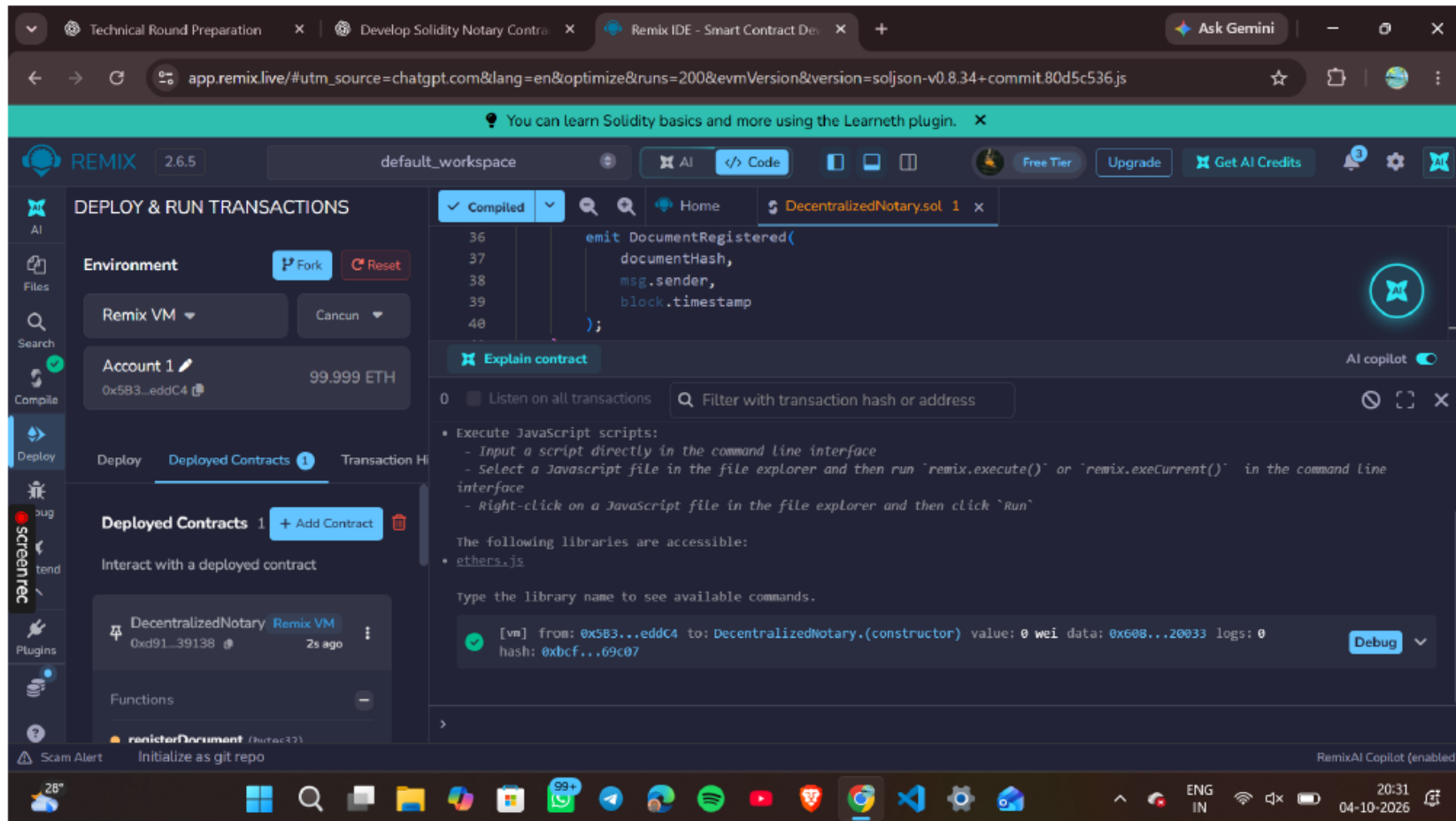
2. Contract Compilation

The DecentralizedNotary contract is successfully compiled using Solidity 0.8.34.



3. Contract Deployment

DecentralizedNotary is successfully deployed in Remix VM (Cancun).



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4. Deployed Contract and Functions

The deployed contract exposes `registerDocument()` and `verifyDocument()`.

The screenshot displays the Remix IDE interface in a web browser. The top navigation bar includes tabs for 'Technical Round Preparation', 'Develop Solidity Notary Contract', and 'Remix IDE - Smart Contract Dev'. The address bar shows the URL: `app.remix.live/#utm_source=chatgpt.com&lang=en&optimize&runs=200&evmVersion&version=soljson-v0.8.34+commit.80d5c536.js`. The main interface is divided into several panels:

- Left Panel:** Contains a sidebar with icons for AI, Files, Search, Compile, Deploy, Debug, and Plugins. The 'Deploy' section is active, showing 'Deployed Contracts' and 'Transaction History'.
- Top Panel:** Displays 'DEPLOY & RUN TRANSACTIONS'. It includes an 'Environment' section with 'Remix VM' and 'Account 1' (0x5B3...eddC4) with a balance of 99.999 ETH. There are buttons for 'Fork', 'Reset', and 'Cancel'.
- Center Panel:** Shows the 'Deployed Contracts' section with a list of contracts. The 'DecentralizedNotary' contract is selected, showing its address (0xd91...39138) and a 'Transact' button. Below it, the 'Functions' section lists the `registerDocument (bytes32)` function with a parameter `documentHash (bytes32)`.
- Right Panel:** Displays the 'DecentralizedNotary.sol' file with the following code:

```
36     emit DocumentRegistered(  
37         documentHash,  
38         msg.sender,  
39         block.timestamp  
40     );  
41 }
```

 Below the code, there is an 'Explain contract' button and a section for 'Listen on all transactions' with a search filter. A list of instructions for using the IDE is provided, including checking transaction details, executing JavaScript scripts, and using the command line interface. A section titled 'The following libraries are accessible:' lists `ethers.js`. A 'Type the library name to see available commands.' input field is also present. At the bottom, a debug console shows a successful transaction: `[vm] from: 0x5B3...eddC4 to: DecentralizedNotary.(constructor) value: 0 wei data: 0x608...28033 logs: 0 hash: 0xbcf...69c07`.

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5. Document Registration – Success

A new document hash is registered successfully.

The screenshot displays the Remix IDE interface with the following components:

- Top Bar:** Includes tabs for 'Technical Round Preparation', 'Develop Solidity Notary Contr...', and 'Remix IDE - Smart Contract Dev...'. The address bar shows 'app.remix.live/#utm_source=chatgpt.com&lang=en&optimize&runs=200&evmVersion&version=soljson-v0.8.34+commit.80d5c536.js'.
- Left Sidebar:** Contains icons for AI, Files, Search, Compile, Deploy, Debug, and Plugins.
- Environment Panel:** Shows 'Remix VM' with 'Account 1' (0x5B3...eddC4) and a balance of '99.999 ETH'. Buttons for 'Fork', 'Reset', and 'Cancel' are visible.
- Deploy Panel:** Displays 'Deployed Contracts' with a table listing 'DecentralizedNotary' (0xd91...39138) deployed '2min ago' on 'Remix VM'. Below the table, the 'registerDocument' function (bytes32) is highlighted, showing a 'Transact' button and input fields for 'Calldata' and 'Params'. The 'verifyDocument' function (bytes32) is also visible with a 'Call' button.
- Code Editor:** Shows the Solidity code for 'DecentralizedNotary.sol' with the following snippet:

```
36     emit DocumentRegistered(  
37         documentHash,  
38         msg.sender,  
39         block.timestamp  
40     );  
41
```
- Transaction History Panel:** Displays a list of transactions. The first transaction is '[vm] from: 0x5B3...eddC4 to: DecentralizedNotary.(constructor) value: 0 wei data: 0x608...20033 logs: 0 hash: 0xbc...69c07'. The second transaction is '[vm] from: 0x5B3...eddC4 to: DecentralizedNotary.registerDocument(bytes32) 0xd91...39138 value: 0 wei data: 0x9b1...11111 logs: 1 hash: 0x152...cd10d'.
- Bottom Bar:** Includes a 'Scam Alert' button, a 'Initialize as git repo' button, and system status information (28°C, ENG IN, 20:33, 04-10-2026).

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6. Document Verification – Success

The registered document returns exists = true, owner address, and timestamp.

The screenshot displays the Remix IDE interface with the following components:

- Top Bar:** Includes tabs for "Technical Round Preparation", "Develop Solidity Notary Contr...", and "Remix IDE - Smart Contract Dev...". The URL bar shows `app.remix.live/#utm_source=chatgpt.com&lang=en&optimize&runs=200&evmVersion&version=soljson-v0.8.34+commit.80d5c536.js`.
- Left Sidebar:** Contains navigation icons for AI, Files, Search, Compile, Deploy, Debug, and Plugins.
- Environment Panel:** Shows "Remix VM" with "Account 1" (address: 0x5B3...eddC4) and a balance of 99.999 ETH.
- Deploy Section:** Includes "Deploy", "Deployed Contracts" (1), and "Transaction History" (2).
- Contract Interaction:** The "verifyDocument" function (bytes32) is called. The results are displayed as follows:
 - 0:bool: exists true
 - 1:address: owner 0x5B38Da6a701c568545dCfcB03FcB875f56beddC4
 - 2:uint256: timestamp 1791126201
- Code Editor:** Shows Solidity code for `DecentralizedNotary.sol`, including the `emit DocumentRegistered` statement.
- AI Copilot:** A panel on the right titled "Explain contract" provides context and lists accessible libraries like `ethers.js`.
- Bottom Bar:** Displays system information including temperature (28°C), search, and taskbar icons, along with the date and time (20:34 04-10-2025).

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7. Duplicate Registration – Rejected

A second attempt with the same hash is rejected with "Document already registered".

The screenshot displays the Remix IDE interface with the following components:

- Top Bar:** Includes tabs for "Technical Round Preparation", "Develop Solidity Notary Contr...", and "Remix IDE - Smart Contract Dev...". The URL bar shows `app.remix.live/#utm_source=chatgpt.com&lang=en&optimize&runs=200&evmVersion&version=soljson-v0.8.34+commit.80d5c536.js`.
- Left Sidebar:** Contains navigation icons for AI, Files, Search, Compile, Deploy, Debug, and Plugins.
- Environment Panel:** Shows "Remix VM" with "Account 1" (0x5B3...eddC4) and a balance of 99.999 ETH.
- Deploy Panel:** Displays "Deployed Contracts" with a list of functions: `registerDocument (bytes32)` and `verifyDocument (bytes32)`. The `registerDocument` function is selected, showing its parameters and a "Transact" button.
- Code Editor:** Shows the Solidity code for `DecentralizedNotary.sol`, including the `emit DocumentRegistered` statement.
- Transaction History:** Shows a transaction from `0xd91...39138` that failed with the message: "Error occurred: revert. Reason provided by the contract: 'Document already registered'".
- Console:** Displays the transaction details and the error message: "Error occurred: revert. Reason provided by the contract: 'Document already registered'".
- Bottom Bar:** Includes a "Scam Alert" button, a "Initialize as git repo" button, and system status information (28°C, ENG IN, 20:35, 04-10-2026).

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8. Unregistered Document Verification

A different hash returns exists = false, zero address, and timestamp 0.

The screenshot displays the Remix IDE interface with the following components:

- Top Bar:** Includes tabs for 'Technical Round Preparation', 'Develop Solidity Notary Contr...', and 'Remix IDE - Smart Contract Dev...'. The URL bar shows 'app.remix.live/#utm_source=chatgpt.com&lang=en&optimize&runs=200&evmVersion&version=soljson-v0.8.34+commit.80d5c536.js'.
- Left Sidebar:** Contains navigation icons for AI, Files, Search, Compile, Deploy, Debug, and Plugins.
- Environment Panel:** Shows 'Remix VM' with 'Account 1' (0x5B3...eddC4) and a balance of 99.999 ETH.
- Deploy Panel:** Displays 'Deployed Contracts' with a contract named 'verifyDocument' (bytes32).
- Code Editor:** Shows Solidity code for 'DecentralizedNotary.sol' with lines 36-41. The code includes an 'emit DocumentRegistered' statement.
- Transaction History:** Shows a failed transaction for 'verifyDocument' with a revert message: 'Error occurred: revert. Reason provided by the contract: "Document already registered."'. It also displays the transaction hash '0x22'.
- Debug Console:** Shows a call log for 'verifyDocument' with parameters: 'data: 0xfe3...22222'.

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9. Transaction History

Remix transaction history shows deployment, successful registration, and the expected failed duplicate attempt.

The screenshot displays the Remix IDE interface. On the left, the 'DEPLOY & RUN TRANSACTIONS' sidebar is active, showing the 'Transaction History' tab. It lists three transactions:

- registerDocument** (tx: 0x071...956bf) - **transact** - 2min ago - **revert** (indicated by a red circle)
- registerDocument** (tx: 0x152...cd10d) - **transact** - 3min ago - **success** (indicated by a green circle)
- DecentralizedNotary** (tx: 0xbcf...69c07) - **deploy** - 6min ago - **success** (indicated by a green circle)

The right-hand pane shows the Solidity code for `DecentralizedNotary.sol` with the following snippet:

```
36     emit DocumentRegistered(  
37         documentHash,  
38         msg.sender,  
39         block.timestamp  
40     );  
41
```

Below the code, the AI Copilot provides an explanation for the revert error:

Explain contract

Reason provided by the contract: "Document already registered".
If the transaction failed for not having enough gas, try increasing the gas limit gently..
Error occurred: revert.
revert
The transaction has been reverted to the initial state.
Reason provided by the contract: "Document already registered".
If the transaction failed for not having enough gas, try increasing the gas limit gently..
If the transaction failed for not having enough gas, try increasing the gas limit gently..

At the bottom, a debug console entry shows a call to `DecentralizedNotary.verifyDocument(bytes32) data: 0xfe3...22222`.

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10. Deployment Transaction Details

Transaction details show the mined status, transaction hash, block hash, and block number.

The screenshot displays the Remix IDE interface with the 'Transaction History' tab selected. The left sidebar shows the 'Deploy' button and the 'Transaction History' tab. The main area displays the 'Transaction History' with three transactions:

- registerDocument** (tx: 0x071...956bf) - **transact** - 3min ago
- registerDocument** (tx: 0x152...cd10d) - **transact** - 4min ago
- DecentralizedNotary** (tx: 0xbcf...69c07) - **deploy** - 7min ago

The right sidebar shows the 'DecentralizedNotary.sol' file with the following code:

```
36 emit DocumentRegistered(  
37     documentHash,  
38     msg.sender,  
39     block.timestamp  
40 );  
41
```

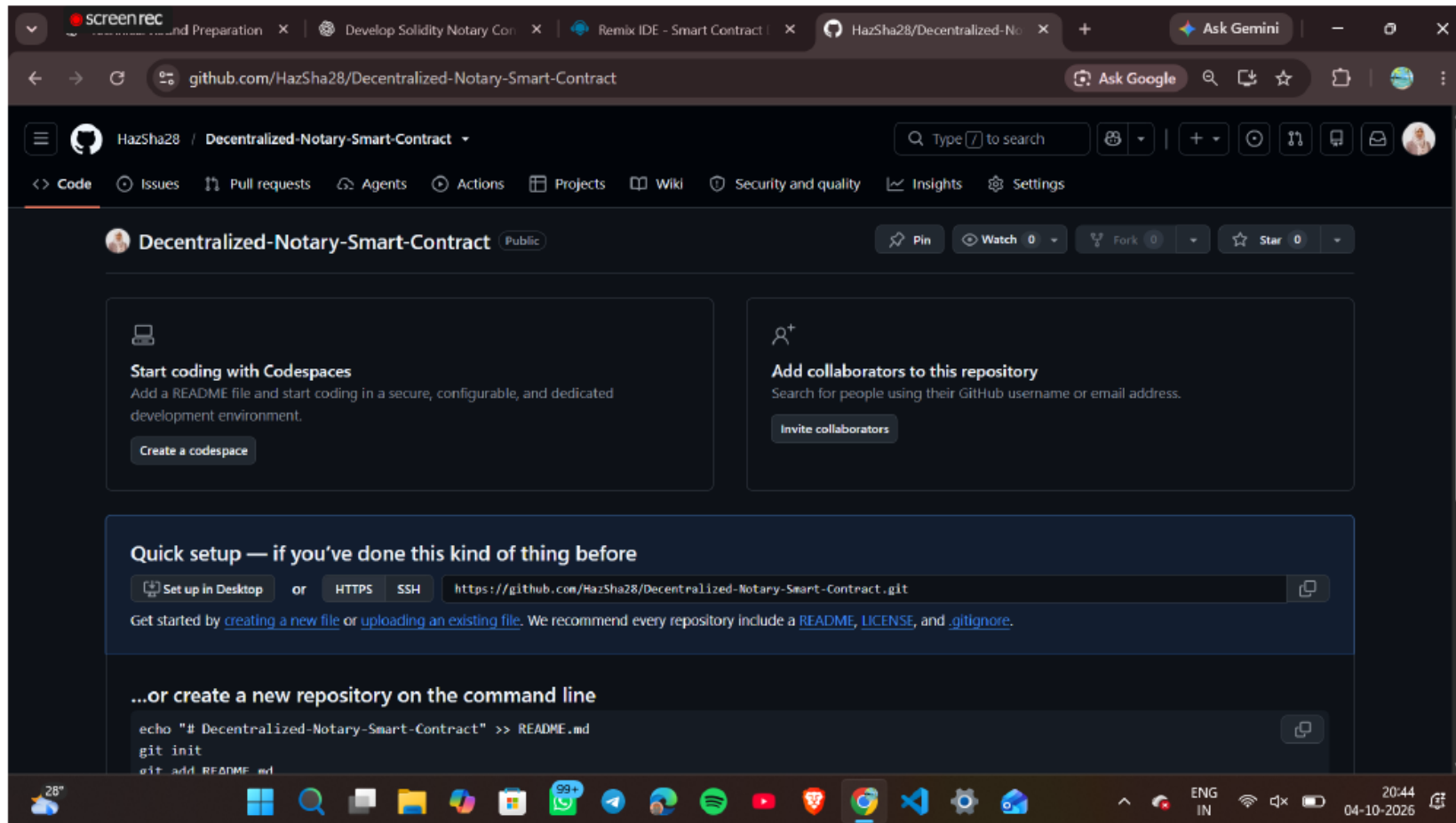
Below the code, the 'Explain contract' section shows the transaction details:

- status**: 1 Transaction mined and execution completed
- transaction hash**: 0xbcf913d3fc42b5810ace0bdc385abb8c26a9d2cdae6a1c36fa03c795a8769c07
- block hash**: 0x89ad0e6e126478996a776e88f73f58ef4176a8b45c31ff0b21fa741ef8f0bd1c
- block number**: 1

The bottom status bar shows the system clock at 20:38 on 04-10-2026.

11. GitHub Repository

The public GitHub repository for the Decentralized Notary Smart Contract.



Test Case Summary

Test Case	Expected Result	Result
Register new document	Document registered successfully	PASS
Verify registered document	exists = true; owner and timestamp returned	PASS
Register duplicate document	Transaction rejected: "Document already registered"	PASS
Verify unregistered document	exists = false; zero address; timestamp = 0	PASS

The screenshots demonstrate successful compilation, deployment, document registration, verification, duplicate protection, and verification of an unregistered document.

DECENTRALIZED NOTARY D-APP

Web application implementation and MetaMask interaction screenshots

Added visual evidence covering wallet connection, transaction confirmation, on-chain registration, and document verification.

127.0.0.1:5500/D-app/index.html

Decentralized Notary

Register and verify digital documents on the blockchain

Connected to MetaMask

Connected: 0x9b88d2b145AeF8DFF9bD8187a4c5b1A816063111

Register Document

Enter the document's SHA-256 hash.

Enter document hash

Register Document

Verify Document

Enter the document hash to verify its ownership.

Enter document hash

Step 1 — Connect MetaMask

MetaMask

Account 1

0x9b88d2...63111

Wallet connected

D-App connected to MetaMask

Decentralized Notary

Register and verify digital documents on the blockchain

Connected to MetaMask

Connected: 0x9b88d2b145AeF8DFF9bD8187a4c5b1A816063111

Register Document

Enter the document's SHA-256 hash.

0xaa

Register Document

Verify Document

Enter the document hash to verify its ownership.

0xaa

Step 3 — Confirm the transaction in MetaMask

MetaMask

Confirm transaction

0 ETH

Network

Local Ethereum

Gas fee

Test ETH

Reject

Confirm

MetaMask transaction confirmation

127.0.0.1:5500/D-app/index.html

Decentralized Notary

Register and verify digital documents on the blockchain

Connected to MetaMask

Connected: 0x9b88d2b145AeF8DFF9bD8187a4c5b1A816063111

Register Document

Enter the document's SHA-256 hash.

0xaa

Register Document

✓ Registration successful! Transaction confirmed.

Verify Document

Enter the document hash to verify its ownership.

0xaa

Step 4 — Hash registered on-chain

MetaMask

Account 1
0x9b88d2...63111

Wallet connected

Document hash registered on-chain

127.0.0.1:5500/D-app/index.html

Decentralized Notary

Register and verify digital documents on the blockchain

Connected to MetaMask

Connected: 0x9b88d2b145AeF8DFF9bD8187a4c5b1A816063111

Register Document

Enter the document's SHA-256 hash.

0xaa

Register Document

✓ Verification successful — document is registered

Verify Document

Enter the document hash to verify its ownership.

0xaa

Step 5 — Verify the same hash

MetaMask

Account 1
0x9b88d2...63111

Wallet connected

Successful document verification